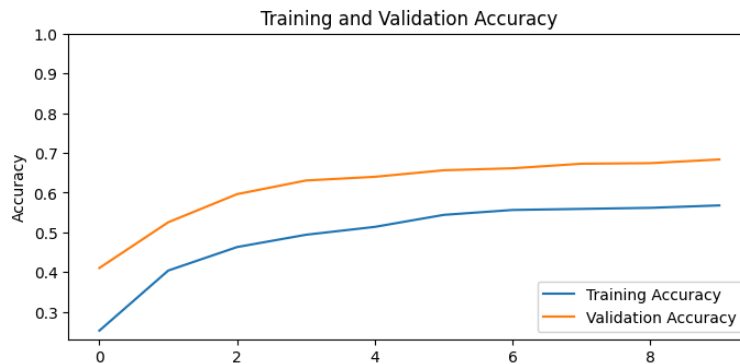


Implementation (Business Focus)

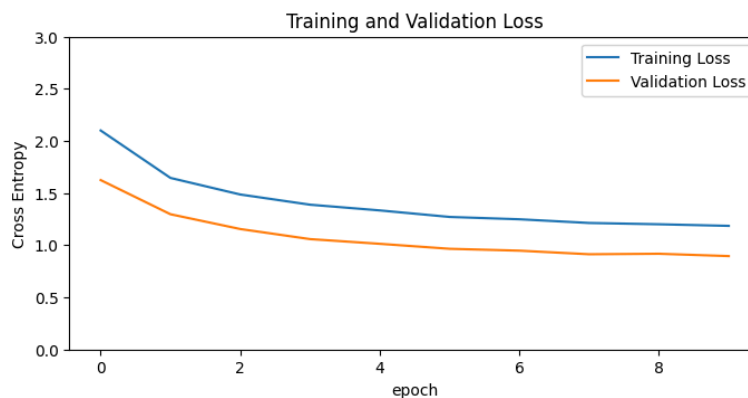
Real Estate Agency X Image Classification Task

- A. Complete the code to become functional according to proposed hints and comments in the code and run successfully.

Google Colab File Link: [Google Colab Notebook](#)



To evaluate model performance, accuracy was used as the primary evaluation metric. Using the CNN base model (InceptionV3) without performing fine-tuning using validation data, initial accuracy was low (0.17) and loss was significant (2.67) for initial performance. After training the model on the validation dataset, the model saw stable improvement to accuracy in both training and validation data. Where the training accuracy was 0.58, while validation accuracy was at 0.68. This gap of 0.1 between the two accuracy metrics suggests moderate overfitting exists in the model. In other words, the model is memorising training data instead of capturing hidden patterns. Which can be addressed through further expanding the dataset or creating more data augmented layers to further improve diversity. Regardless, this is a common issue in CNN models.



Without a testing subset, final results of model performance were essentially tested on the validation dataset, showing an accuracy of 0.684 & loss of 0.896 as mentioned earlier. Where the model's validation loss drops approximately 45% from 1.625 to 0.896, displaying consistent gains in each epoch iteration. Furthermore, the model trend also continues to improve marginally by 10 epochs, where additional training epochs could yield further improvement to performance in both accuracy and loss to a certain extent. Hence, the performance of the model presents solid results for a property image classification model, but could advance further with additional epochs and data.

B. Explain and present the results with respect to success criteria and business objectives achievement.



To understand model success, results will be evaluated against and discussed in the context of the success criteria and business objectives defined in the project proposal.

ML Success Criteria: Although F1-Score ($\geq 70\%$) was defined as the metric for evaluation previously due to imbalance dataset consideration, accuracy was employed as the primary reported metric instead. As the dataset had a balanced distribution in each class, the validation accuracy of 0.684 is typically close to the average precision and recall value (Google ML Edu, n.d.). As a result the target ($\geq 70\%$) of F1-Score becomes numerically similar to accuracy, in which the validation accuracy falls slightly under after 10 training epochs. But with additional training time and epochs, it displays the potential to meet the defined classification metric target.

Business Objectives: In terms objectives, the classification model results was able to meet each objective to a certain extent as explained in the following:

1. **Improve Employee Satisfaction** - Using the visualisation above as random classification samples, the model was able to fulfill the classification task of categorising property photos with 68% accuracy. Although the task of image classification cannot be fully automated due to moderate accuracy, the implementation of the current model should at least lower the workload and time of current staff. Where the output of target and true label should be constantly monitored by staff members, which removes manual repetitive work.
 2. **Reduce Operational Costs** - Through employing the current model, though cannot be fully automated, the number of labour and resources required for the task can be reduced. It is recommended that the Agency implement the model under strict monitoring by workers to avoid miscategorisation of photos, as demonstrated in the sample that inaccurate prediction frequently occurs in the current model.
 3. **Reallocate Resources for Growth** - Despite the necessary need for staff to consistently monitor outputs and fix miscategorisation as it occurs, the model is still able to remove a majority of the time (16,650 hrs) and resources required for the manual classification process. By reducing resources and labour, staff members can engage in more meaningful tasks to build value and to encourage future organisational growth.
- C. Explain future improvements/features for the built product and propose related new products.

Dataset Expansion: To further improve model results, it is recommended to continue model training to maximise performance using an extended dataset with balanced class distribution, and increase the number of epochs (e.g. 15 - 20). Furthermore, supplementary samples can be provided by adding more data augmentation layers in addition to horizontal flipping. Including other transformations such as Gaussian noise and filters to get a more generalised and more accurate model.

Hardware Improvement: To mitigate overfitting issues, increased data is required for model training which leads to greater demand in computational resources. The current classification model was trained on a low spec laptop with limited GPU and RAM power, which required approximately 2 hours to complete training. For that reason, to account for larger datasets and to reduce overall model training time in the foreseeable future, a desktop equipped with moderately powerful GPU and RAM is suggested for bottleneck reduction as dataset expands and number of epoch increases in training.

New Related Product Proposal: In extension to the property photo classification tool, a complementary product that could be considered is property condition classification using

ML solutions. This tool would automatically analyse the interior and exterior images from a property to classify the condition of the property based on images compiled.

Property condition/type labels could assign labels such as “new”, “recently renovated”, “established home” to provide property insights to potential buyers. It could also identify property types including apartment, studio, and house. When paired with the existing photo classification tool, other tedious tasks performed by workers can be further removed by automating small and simple tasks such as condition classification.

As this would also employ property photos for training, the same dataset can be utilized through data preprocessing, which is a realistic product for a company only starting to implement ML solutions into the operation environment.

Follow Up Questions (Property Description Generation)

A. What Machine Learning techniques are suitable and elaborate why?

In addition to image classification, Real Estate Agency X has displayed interests in automating another tedious task such as writing property descriptions for clients’ listing properties. However, in contrast to image classification, CNN models won’t be suitable for text based tasks such as description generation. The company should instead incorporate a recurrent neural network (RNN) model such as LSTM for generating property description.

Unlike other neural networks, RNN contains an internal loop within the network which allows it to carry information forward from previous steps and update its state as it processes through sequences. In other words, the model comprehends the context and structure of texts better than other ML techniques as it captures the context of the sequence and understands the dependencies between texts. As a result, it is particularly useful for language modelling such as description generation in this context.

B. What are the main requirements and challenges for the successful development of your proposed solution?

In order to build a successful language model for property description, the following requirements and challenges should be considered:

- **Requirements:** To build another model, a new dataset is necessary for model training, validation, and testing. This new dataset should contain property images paired with text descriptions, with around 10,000 - 15,000 property samples. In addition, the data included should derive from a range of property types (e.g. apartments, houses, etc.) with balanced distribution.

- Challenges: One of the primary issues to consider for this tool is the quality and compliance of the generated description. The output of the model could generate bland and unappealing descriptions leading to misrepresentation of the client's property and damaged reputation. Furthermore, inaccurate or misleading description of property characteristics could risk breaching the Australian Consumer Law. For that reason, the accuracy and quality of AI-generated property description is a critical consideration for this proposed product.
- C. How the output of your proposed solution for the previous problem (Property Photo Classification) can help your solution for this problem (Property Description Generation) if it can?

The outputs generated from the property photo classification tool can help and introduce enhanced quality and accuracy if incorporated into the RNN model through the labels attached to each image (predicted by photo classification tool). This will greatly reduce the time spent labeling images for the description dataset, and enable the model to call on specific room category descriptions based on image labels. In addition, it will also act as an indicator of which categories to include within the description. For instance, the text could directly refer to each category (e.g. "Spacious master bedroom", "Large open kitchen", etc.) as a result of the contextual output provided by prior classification. This essentially connects the gap between raw images and contextual labels for more engaging property description, and streamlines the classification process of property images which will be a required process for the property description generation tool.

Reference

Google Machine Learning Education (n.d.). ML Concepts: Classification: Accuracy, recall, precision, and related metrics.

<https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall>